



MANIPAL ACADEMY OF HIGHER EDUCATION

Manipal Institute of Technology, Bengaluru

BTech VI Semester

End Semester Examination - April / May -2025

HUMAN COMPUTER INTERFACE [CSE- 4069]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Discuss the causes of human error in HCI. How can designers reduce error through interface design? (5)
 - A)
 - B) Discuss the characteristics of devices used for pointing and drawing. How does Fitts' Law apply to pointing and positioning tasks? (5)
- 2) Use Norman's Execution-Evaluation cycle to analyze a user setting a timer on a microwave oven. Explain the complete steps of the model. (5)
 - A)
 - B) You are tasked with designing an interface for an in-car infotainment system that minimizes driver distraction. An infotainment system is a central digital system in vehicles that combines information and entertainment features. Apply appropriate interaction paradigms to propose a solution and justify your choices. (5)
- 3) Explain the key principles of screen layout that support better usability. (5)
 - A)
 - B) Given an ATM interface, explain how compliance with interface standards would improve its accessibility and consistency. (5)
- 4) Apply appropriate expert analysis technique to evaluate the interface of a hospital patient record system. Describe the techniques, the steps involved and expected outcomes. (5)
 - A)
 - B) Why user support is required in HCI systems? Identify any four approaches that can be used to provide user support in online ticket reservation system. Provide appropriate reasons for the methods identified. (5)
- 5) What is task analysis? Give one example. Compare and contrast task analysis with system analysis and cognitive models. (5)
 - A)
 - B) A student information system needs to be developed for a college. Apply the CUSTOM approach to capture the requirements of the stakeholders of this system. Explain each step in detail. (5)

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